Falling into orbit

A fall with a sideways start

The launch gives the ball sideways speed. Gravity keeps turning its velocity toward Earth. With a slow launch, the ground gets in the way.

Faster sends it farther

A faster launch carries the ball farther around the curved surface before it lands. It is still falling the whole time.

Keep falling. Keep missing.

At circular speed, the path bends just as quickly as the surface curves away. The ball stays at one radius because it is continually falling around Earth.

Falling into orbit

An orbit need not be a circle

A little more speed lifts the far side of the orbit. Gravity slows the ball as it climbs and speeds it up as it comes back.

Enough energy to leave

At the escape threshold, the ball can keep moving outward without another push. That threshold is sqrt(2) times the circular speed at the same launch height.

Model assumptions

A spherical Earth and an unpowered test mass. No atmosphere, rotation, other bodies or relativity. Launch height is 5% of the radius. Speeds are relative to circular speed at that height. The companion is original instructional material, not a page from Feynman.